



Bio-Manufacturing of Engineered Living Materials for Martian Construction: Design of the Synthetic Community

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As the next step in extraterrestrial exploration, many engineers and scientists revealed their intense interest in enabling multiplanetary human life, including colonizing Mars. This study demonstrates that architecture on Mars can be realized by designing a synthetic community, including diazotrophic cyanobacteria and filamentous fungi, which produce large amounts of biomaterials to bond Martian regolith particles into a consolidated body. Through 3D printing, a wide range of structures can be fabricated, such as buildings, houses, tables, and chairs. Within the synthetic community, diazotrophic cyanobacteria will (1) fix carbon dioxide and dinitrogen from the atmosphere and convert them into oxygen and organic nutrients to help the survival and growth of filamentous fungi and (2) increase the concentration of carbonate ions by photosynthetic activities. Filamentous fungi will (1) bind metal ions onto fungal cell walls and serve as nucleation sites for biomineral production and (2) enhance the growth of cyanobacteria by providing them with water, minerals, and carbon dioxide. In this study, such coculture systems have been created and displayed robust growth solely based on Martian regolith simulants, air, light, and an inorganic liquid medium without any additional carbon or nitrogen sources. The cyanobacterial and fungal growth in such coculture systems is much more robust than their axenic growth due to mutual interactions. The amounts and morphologies of the precipitated crystals vary remarkably depending on the cultivation condition.
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Keywords: engineered living material, construction material, Martian construction, synthetic biology, mutualistic coculture, advanced materials and processing, design for manufacturing

1 Introduction

As the next step in extraterrestrial exploration, many engineers and scientists revealed their intense interest in enabling multiplanetary human life, including colonizing Mars. Without the presence of any antecedent, the first Martian architecture is typically portrayed as exquisite industrial prefabricates, as shown in Fig. 1(a) [1]. However, the challenges of reaching such a distant realm, transporting entire sets of building parts, and assembling and maintaining them in environments that are far more challenging than even the most extreme environments on Earth are well beyond our current science and technology.

The goal of this project is to bring in game changers—radical changes to the state-of-the-art construction technologies such that the construction process can adjust to the restricted resources and adverse environments on new planets. The main progress of the project is summarized here, which aims to engineer microorganisms

to create biomaterials that glue Martian regolith particles into structures, as shown in Fig. 1(b) [2].

A two-component synthetic lichen system is created, as shown in Fig. 2(a). Diazotrophic cyanobacteria oversee (1) fixing carbon dioxide and dinitrogen from the atmosphere and converting them into oxygen and organic nutrients to help the growth of filamentous fungi, as shown in Fig. 2(b), and (2) increasing the concentration of carbonate ions by photosynthetic activities, which is important for biomineral precipitation. Filamentous fungi oversee (1) attracting metal ions onto fungal cell walls and serving as nucleation sites for biomineral production and (2) enhancing the growth of cyanobacteria by providing them with water, minerals, and carbon dioxide. Both components secrete biopolymers that enhance the adhesion and cohesion among Martian regolith and precipitated particles.

2 Background and Motivation

To realize in situ utilization of Martian regolith, various methodologies for bonding regolith particles have been proposed. Scott et al. [3] bonded Martian regolith simulants using a magnesium-

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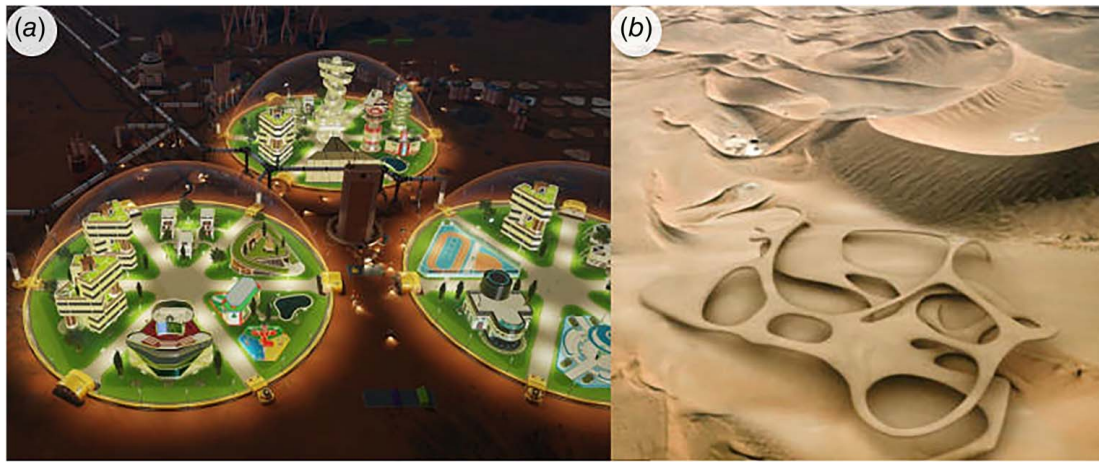


Fig. 1 (a) A Google image search for “Mars colony” returns a lot of hi-tech structures [1]. **(b)** This study proposes that microorganisms can be engineered to create abundant biomaterials that will glue Martian regolith particles into biostructures [2].

based cement. It has been found that magnesium oxide, which needs to be extracted from magnesium carbonates on Mars, can react with amorphous silica to form magnesium-silica-hydrate, which functions like the calcium-silicate-hydrate formed in ordinary Portland cement. Wan et al. [4] mixed molten sulfur with Martian regolith simulants to produce sulfur concrete. Sulfur needs to be extracted from gypsum or pyrite minerals on Mars. Alexiadis et al. [5] created and tested geopolymer concrete using Martian regolith simulants. Geopolymerization hinges on the activation of regolith using a variety of activators and thus its feasibility requires the production of activator materials on Mars. In summary, all these approaches require significant amounts of human assistance, and thus cannot combat the obvious challenge of manpower shortage on Mars.

On the other hand, microbe-mediated self-growing technology has been intensely investigated during the past decades. Dosier et al. utilized bacterial biomineralization to consolidate sand particles into masonry units [6]. She founded a company called bioMA-SON. Besides biominerals, fungal mycelium has also been utilized as bonding materials. A company called Ecovative Design was founded based on its patented process [7]. Another company, MycoWorks, also uses fungal mycelium to create building blocks and leather-like fabrics.

Recognizing the significant potential of self-growing technology in space exploration and colonization, the Indian Institute of Science demonstrated the feasibility of using ureolytic bacteria to promote the production of calcium carbonate to consolidate Martian regolith simulants into bricks [8]. In the United States, NASA Ames Research Center has an ongoing project entitled “Myco-Architecture off Planet,” which explores the feasibility of using fungal mycelium to bond Martian regolith simulants [9].

As a very young but promising technology, microbe-mediated self-growing technology has reached a significant degree of advancement. However, it still has many drawbacks. None of the current practices is completely autonomous, because the microbes used in the existing studies are limited to a single species or strain, especially ureolytic bacteria, and thus, their survivability requires a continuous nutrient supply. Many researchers seriously investigated the effect of different nutrient dosing methods on the microbial functionality over time [10].

This project aims to research a completely autonomous self-growing technology by designing a synthetic community and making use of the advantages of multiple species. This system eliminates the need for external nutrient supplies, which is of utmost significance for space construction.

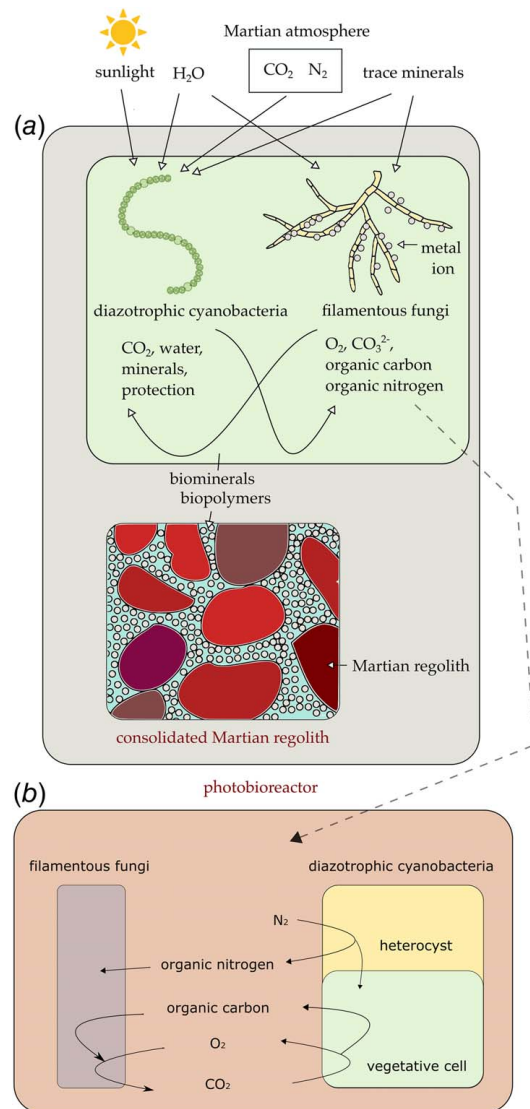


Fig. 2 (a) Schematic illustration of the self-sustaining system. **(b)** Diazotrophic cyanobacteria fix carbon dioxide and dinitrogen from the atmosphere and convert them into oxygen and organic nutrients to assist the growth of filamentous fungi.



Fig. 3 (a) Field collection at Nottingham County Park. (b) Soil samples were collected and tested for pH. (c) For each site, healthy plant individuals were collected and kept on ice during transportation.

3 Design Considerations of the Synthetic Community

In this project, heterotrophic filamentous fungi, instead of heterotrophic bacteria, are selected to fit the role of bonding material producers. The design consideration is that filamentous fungi are better candidates than bacteria as bonding material producers due to their superior survivability against harsh conditions and excellent ability to promote large amounts of biominerals. When investigating microbe-based self-healing concrete, ureolytic bacteria have been used as healing agents for several decades, but recent research demonstrated the advantages of using fungi as biomineral producers [11]. Some researchers observed that fungi-based healing time could reach 11.3% of the time required by bacteria [12]. Some studies demonstrated that filamentous fungi could promote much larger amounts of calcium carbonate precipitates within the same time duration compared with bacteria [13].

The second design consideration is that a synthetic lichen system can be created by pairing heterotrophic filamentous fungi with photoautotrophic diazotrophic cyanobacteria that exhibit mutualistic interactions. Some proof-of-concept studies [14] constructed such self-supporting synthetic systems and observed that the phototroph was able to provide organic nutrients for the heterotroph whereas the heterotroph enhanced the growth of the phototroph. Note that none of the previous studies were conducted in the environment of Martian regolith simulants.

4 Materials and Methods

4.1 Collection of Alkaliphilic Strains. In the past years, new Martian regolith simulants kept being developed, among which MGS-1 was fabricated based on individual components to match the mineralogy of Martian regolith, making it so far the most accurate basaltic Martian regolith simulant. The pH of MGS-1, being approximately 9.5 [15], remains the most significant challenge for this project since this high pH inhibits the growth of most microorganisms. Thus, alkaliphilic cyanobacteria and fungi need to be obtained from natural environments or via genetic editing.

Wild-type isolates typically possess exceptional resistance to temperature fluctuations and superior adaptation to nutrient-poor and dry habitats. In the past few years, grass root-associated fungi were collected at the Nottingham Serpentine Barrens in Pennsylvania, the Pine Barrens in New Jersey, and the Rocky Mountains of Alberta in Canada, respectively, where the soils have high pH and low nutrients. Figure 3 shows the field collection conducted at Nottingham County Park, which is part of the Nottingham Serpentine Barrens.

Some wild-type isolates are found to possess a superior ability to promote large amounts of precipitates, but they do not possess sufficient survivability against the alkaline environment of MGS-1. Genetic editing can be utilized to address this issue. For filamentous fungi, alkaline pH is registered by a signal transduction pathway involving the products of at least six *pal* genes and transmitted to a transcription factor encoded by the *pacC* regulatory gene. The *pacC* mutations in the *pacC* regulatory gene lead to alkalinity mimicry [16].

Nonphotosynthetic bacteria survive high pH using a wide range of mechanisms, among which the best-studied approach is the increased expression and activity of monovalent cation/proton antiporters [17]. These transmembrane proteins regulate the intracellular pH via the uptake of protons. A variety of such antiporters are recognized in cyanobacteria [18–20]. Their role in pH homeostasis can be investigated by gene knockout studies.

4.2 Survivability and Mutualism Testing. As shown in Fig. 4, the investigation is conducted in four steps. The collected fungal strains first go through survivability testing. The strains that demonstrate optimal survivability in the growth medium containing Martian regolith simulants are selected for the second testing, i.e., the mutualism testing. A series of coculture systems are created, and their growth is solely based on Martian regolith simulants, air, light, and an inorganic liquid medium, i.e., a modified Bold's basal medium [21], with or without additional carbon or nitrogen sources is observed. For each monoculture or coculture, four different cases are tested, i.e., C + N+, C + N–, C – N+, and C – N–.

To assess microbial growth in each well, three quantitative characterization methods are applied, i.e., resazurin assay, fungal plating on selective medium, and phycocyanin assay. The coculture systems with the optimal survivability and mutualism without any additional carbon or nitrogen sources are selected as candidates for Martian construction. The detailed experimental protocols of resazurin assay, fungal plating on selective medium, and phycocyanin assay can be found in the previous publication [21].

Neither cyanobacteria nor fungi could directly thrive on Mars. The Martian atmosphere has a pressure of 7.5 mbar, only 0.7% of that of Earth, which does not support the metabolism of most microbes. Since there exists only 2.8% dinitrogen in the Martian atmosphere, cyanobacteria cannot extract enough nitrogen to support their diazotrophic growth. In addition, the surface temperature on Mars shows extreme diurnal fluctuations from 174 K to 295 K, unsuitable for microbial growth. Finally, the absence of an ozone shield means that radiation protection is needed. Hence, a

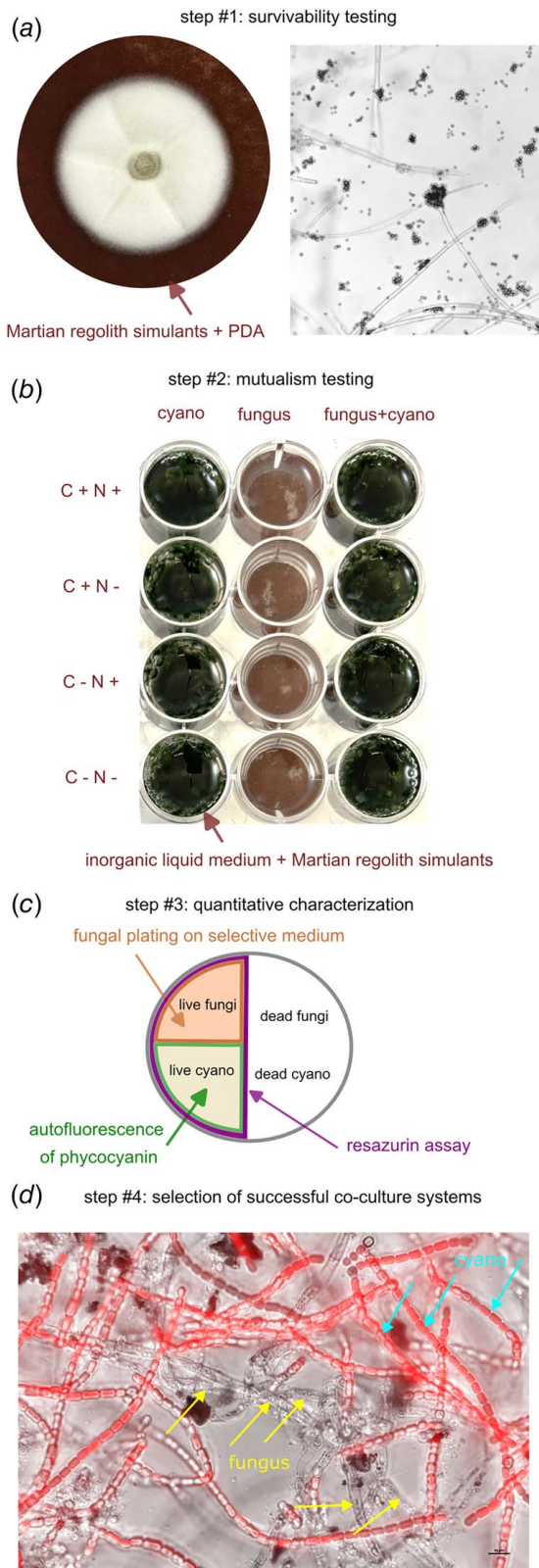


Fig. 4 The investigation is conducted in four steps

photobioreactor is needed to provide the microbes with protection against damaging radiation and optimal growth conditions. The potential engineering challenges and solutions for deploying photobioreactors on Mars have been addressed in several studies [22,23].

Survivability and mutualism testing are performed within a laboratory-scale photobioreactor, as shown in Fig. 5(a). Except

for the small quantities of cyanobacterial and fungal spores, all the inputs for the photobioreactor can be directly found on Mars, i.e., water, carbon dioxide, dinitrogen, and Martian regolith. Some coculture systems demonstrate robust growth solely based on the presence of light, water, carbon dioxide, dinitrogen, and trace minerals extracted from Martian regolith simulants, as shown in Fig. 5(b).

4.3 Sugar Detection During Coculturing. To understand if the cyanobacterial component provides nutrients for its fungal partner in the C – N– cocultures, metabolomics is an essential tool to study the differences in metabolites between monocultures and cocultures. Therefore, the metabolites from monocultures and cocultures, both grown solely on Martian regolith simulants, air, light, and an inorganic liquid medium without additional carbon or nitrogen sources, are of utmost importance for investigation.

In this study, Benedict’s test is conducted as a preliminary analysis to detect reducing sugars, which are critical for fungal survival under oligotrophic conditions. To perform the test, 1 mL of supernatant is collected from each monoculture and coculture well and mixed with 2 mL of Benedict’s reagent. Glucose, a known reducing sugar, is used as a control at varying concentrations. The mixtures are vortexed and then boiled in a water bath for 5 min to facilitate the reduction reaction. During this process, any reducing sugars present in the supernatant reacted with copper (II) ions in Benedict’s reagent, resulting in a visible color change. The degree of the color change from blue to green, yellow, orange, or brown depends on the concentration of reducing sugars present in the supernatant of each well. After cooling, the samples’ optical density is measured at 600 nm using a spectrophotometer to quantify the color change.

4.4 Material Characterization of Biomaterials. Due to the complexity of the biomineralization process, material characterization technologies are employed to understand the effect of various influencing factors on the production of biomaterials, such as fungal strain/species, cyanobacterial strain/species, atmospheric composition, illumination cycle, temperature range, cultivation duration, and growth medium composition.

To monitor the structural and morphological changes of the solid precipitates over time, scanning electron microscopy (SEM) and energy dispersive X-ray spectroscopy (EDS) are performed at the early, intermediate, and terminal stages of each mutualism test. SEM and EDS provide a three-dimensional overview of the morphological and composition evolution of the solid precipitates.

To test the samples in each well, the growth medium is removed from the culture and the samples are washed by immersing twice in 70% ethanol for 15 min. Representative samples are then spread as a thin layer on cover slips and left overnight to dehydrate with 95% ethanol. The samples are then completely dried in an oven at 40 °C for 2 days. The dried samples are mounted onto aluminum stubs using double-sided carbon tape and sputter-coated with a 5- to 10-nm layer of chromium to ensure electrical conductivity. The crystals are observed using a Hitachi S4700 Field-Emission SEM and an FEI Helios NanoLab 660 SEM at accelerating voltages between 5 kV and 20 kV. EDS is performed by an EDAX Octane Elect EDS system.

5 Results

5.1 Survivability and Mutualism Results. Through the four-step investigation, it can be concluded that the coculture systems that demonstrate optimal survivability and mutualism without any additional carbon or nitrogen sources include six stable symbioses consisting of a diazotrophic cyanobacterium *Anabaena* sp. (UTEX2576) and a filamentous fungus *Penicillium sipitatus* (ATCC10500), *Aspergillus niger* (ATCC16888), *Trichoderma reesei* (ATCC13631), *Trichoderma viride* (HF3TV), *Trichoderma viride* (HF3MP), or *Aspergillus flavus* (ATCC9643). Their

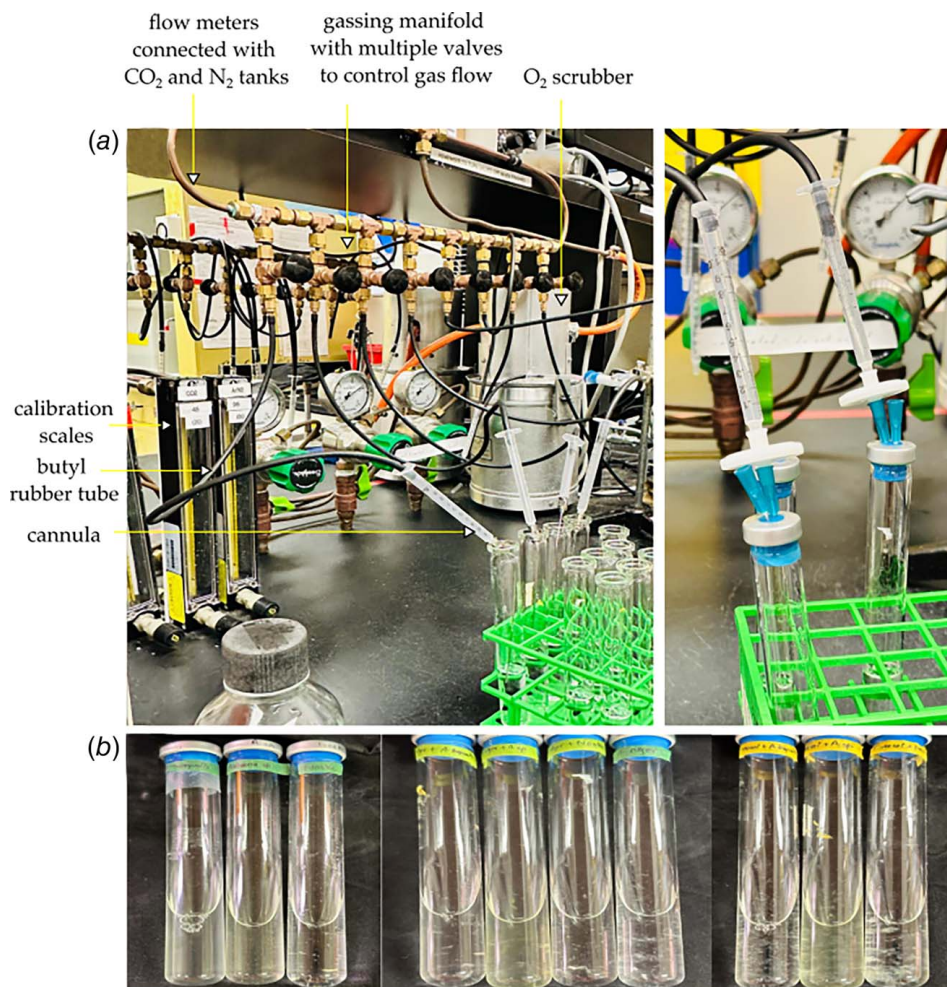


Fig. 5 (a) Survivability testing is performed using a laboratory-scale photobioreactor. (b) Some coculture systems demonstrate robust growth solely based on the presence of light, water, carbon dioxide, dinitrogen, and trace minerals extracted from Martian regolith simulants.

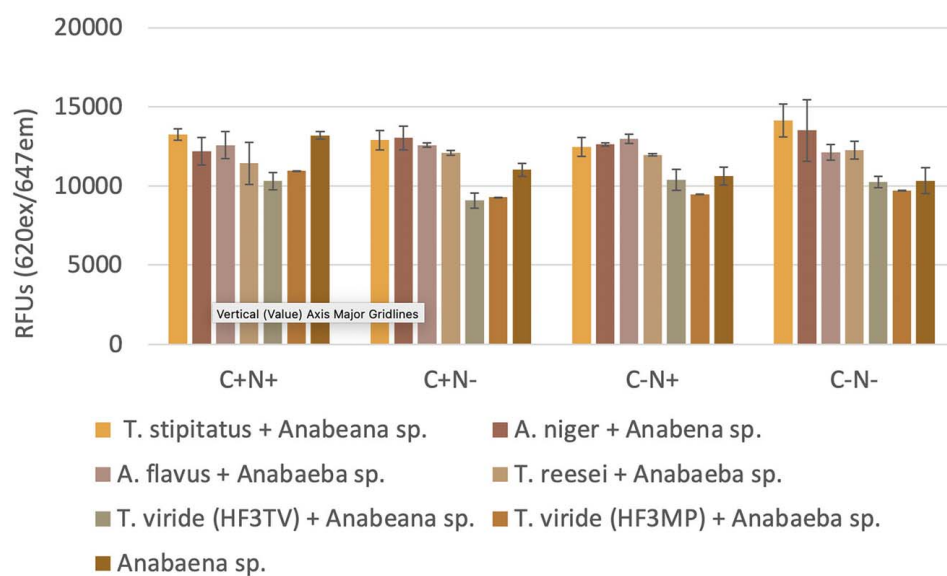


Fig. 6 Results of phycocyanin autofluorescence measurement after 45-day incubation, characterizing the number of live cyanobacterial cells. All data are averages of three independent experiments, and the error bars are the standard deviations.

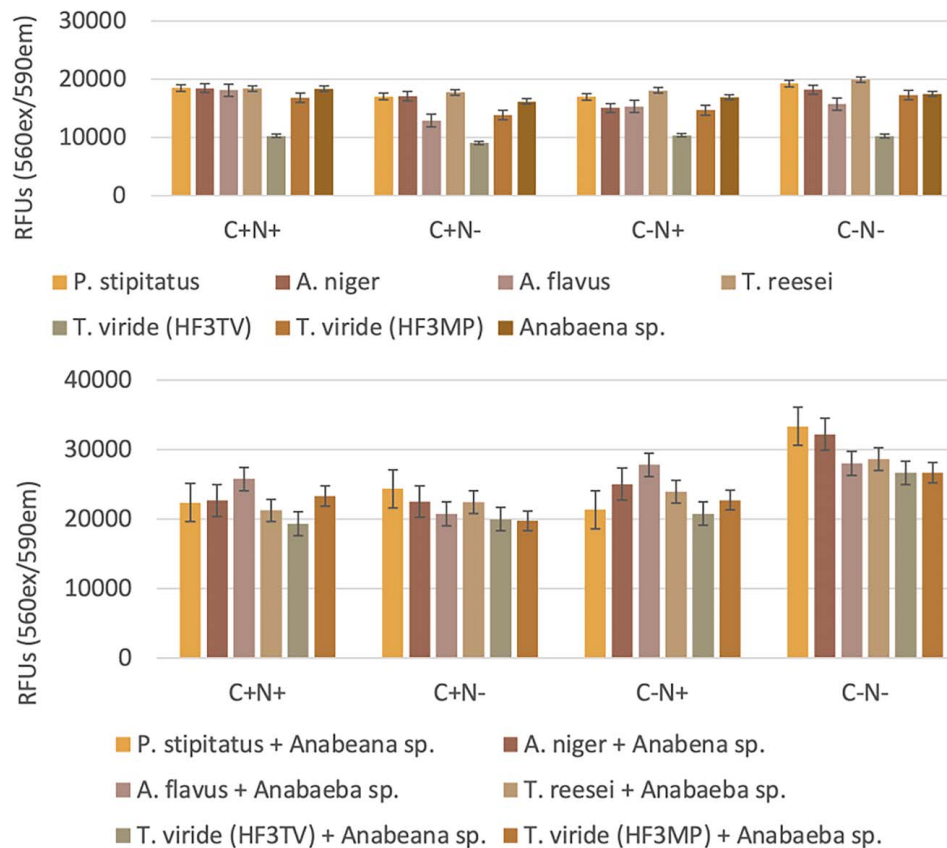


Fig. 7 Results of resazurin assay after 45-day incubation, characterizing the number of live cells, both fungal and cyanobacterial. All data are averages of three independent experiments, and the error bars are the standard deviations.

successful growth on Martian regolith simulants, air, light, and an inorganic liquid medium without any additional carbon or nitrogen sources has been tested and confirmed. In addition, based on the testing results within the photobioreactor, the coculture of

Anabaena sp. and *A. niger* and the coculture of *Anabaena* sp. and *T. reesei* demonstrate robust growth solely based on the presence of light, water, carbon dioxide, dinitrogen, and trace minerals extracted from Martian regolith simulants.

The autofluorescence of phycocyanin characterizes the number of live cyanobacterial cells, and the results after the 45-day incubation period are shown in Fig. 6. Note that the cocultures are made by mixing the two components in equal proportion, and thus, initially, they had only one-half of the live cyanobacterial cells in comparison with the monoculture of cyanobacteria. Therefore, in the coculture wells (the first six cases in Fig. 6), the cyanobacteria generally displayed a much faster growth rate than those in the cyanobacterial monoculture (the last case in Fig. 6).

The resazurin assay assesses the number of live cells, both fungal and cyanobacterial, and the results after the 45-day incubation period are shown in Fig. 7. It can be clearly seen that the cyanobacterial or fungal growth in the coculture systems is much better than their axenic growth, demonstrating the importance of mutual interactions.

The method of fungal plating on a selective medium assesses the number of live fungal cells in each well, and the results after the 45-day incubation period are shown in Fig. 8, indicating that there are significant amounts of live fungal cells after 45 days of incubation, even when no supplemented carbon or nitrogen source is present. Note that this method can only provide a rough estimation of the live fungal cells and a more accurate method will be explored in future work.

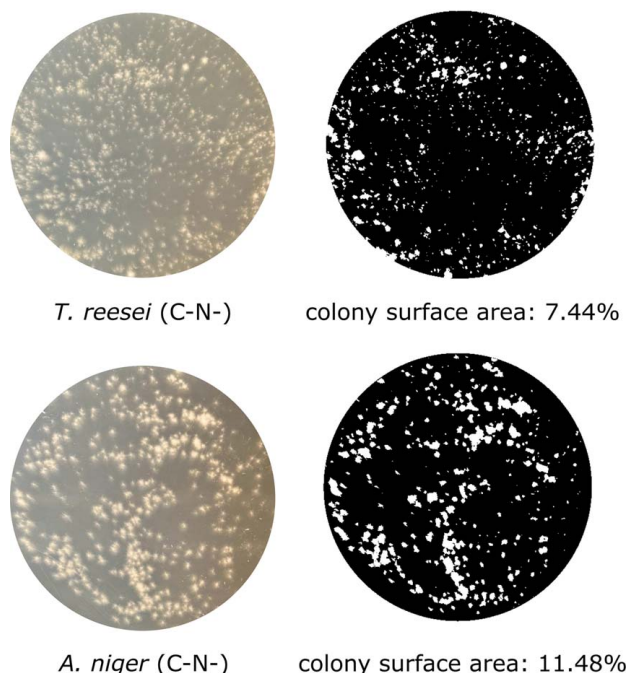


Fig. 8 Results of fungal plating on selective medium after 45-day incubation, assessing the amount of live fungal cells

5.2 Sugar Detection During Coculturing. Benedict's test is used to identify reducing sugars in the supernatant of monoculture and coculture wells. The degree of the color change reflects the percentage of reducing sugars present in the supernatant of each well,

as shown in Fig. 9(a). The change in color in the coculture supernatants indicates that the cyanobacteria are producing reducing sugars, not initially provided in the media, for its fungal partner, as shown in Fig. 9(b).

However, Benedict's test is limited to the detection of reducing sugars. *Anabaena* is known to produce not only reducing sugars but also nonreducing sugars, amino acids, organic acids, and fatty acids, all of which could support fungal growth. Therefore, to fully understand the metabolic interactions between cyanobacteria and fungi, it would be valuable to employ advanced technologies such as untargeted metabolomics, which would provide deeper insight into how metabolic exchanges facilitate their symbiotic association. But this work will be published separately.

5.3 Characterization Results of Biomaterials. It has been observed that the cultivation condition could significantly influence the precipitation process, and even under the same cultivation condition, the amounts and morphologies of the precipitated crystals vary remarkably depending on the fungal component, as shown in Fig. 10. When the cyanobacterial component is fixed to be *Anabaena* sp., *T. reesei* precipitates prism-like crystals, *T. viride* (HF3MP) precipitates plate-like crystals, *T. viride* (HF3TV) precipitates rod-like crystals, *P. sipitatus* precipitates crystals with gel-like structures, and on the other hand, *A. niger* does not precipitate large amounts of crystals, but its mycelial network shows notable growth, which is more suitable for mycelium-based self-growing technology such as “Myco-Architecture off Planet” [9]. EDS analysis showed that the crystals are mainly calcium carbonate. In the microbe-free control medium, no crystal growth is observed.

6 Discussion

This study demonstrates the feasibility of constructing synthetic symbioses that can successfully grow on Martian regolith simulants, air, light, and an inorganic liquid medium without any additional carbon or nitrogen sources. Within the synthetic community, the diazotrophic cyanobacteria synthesize carbohydrates from carbon dioxide, satisfying the community's energy and carbon requirements, and convert atmospheric dinitrogen into organic nitrogen, satisfying the community's nitrogen requirements for essential cellular processes. The filamentous fungi act as metal ion attractors that deposit biomineral precipitates.

The use of Martian regolith simulants is a significant limitation of this study. While simulants offer a way to study the potential of microbial growth on Mars without the need for expensive real samples, they may not perfectly replicate the complexities of real Martian regolith. Martian regolith is diverse, and simulants struggle to fully capture the effects of the harsh environment on Mars and can only represent a simplified version of its mineralogy, chemistry, and particle size distribution. In addition, the average gravitational acceleration on Mars is about 38% of the gravity of Earth and the impact of the reduced gravity on biomineralization is unclear. Note that this study aims to use biominerals to glue regolith particles into a cohesive body, whereas the project entitled “Myco-Architecture off Planet” explores the feasibility of using fungal mycelium to bond regolith particles [9]. The mechanical strength and durability of the final products are expected to be different.

7 Conclusion and Future Work

In this study, synthetic symbioses consisting of diazotrophic cyanobacteria and filamentous fungi were constructed and their successful growth on Martian regolith simulants, air, light, and an inorganic liquid medium without any additional carbon or nitrogen sources has been tested and confirmed. It has been observed that the cultivation condition could significantly influence the precipitation process, and even under the same cultivation condition, the amounts

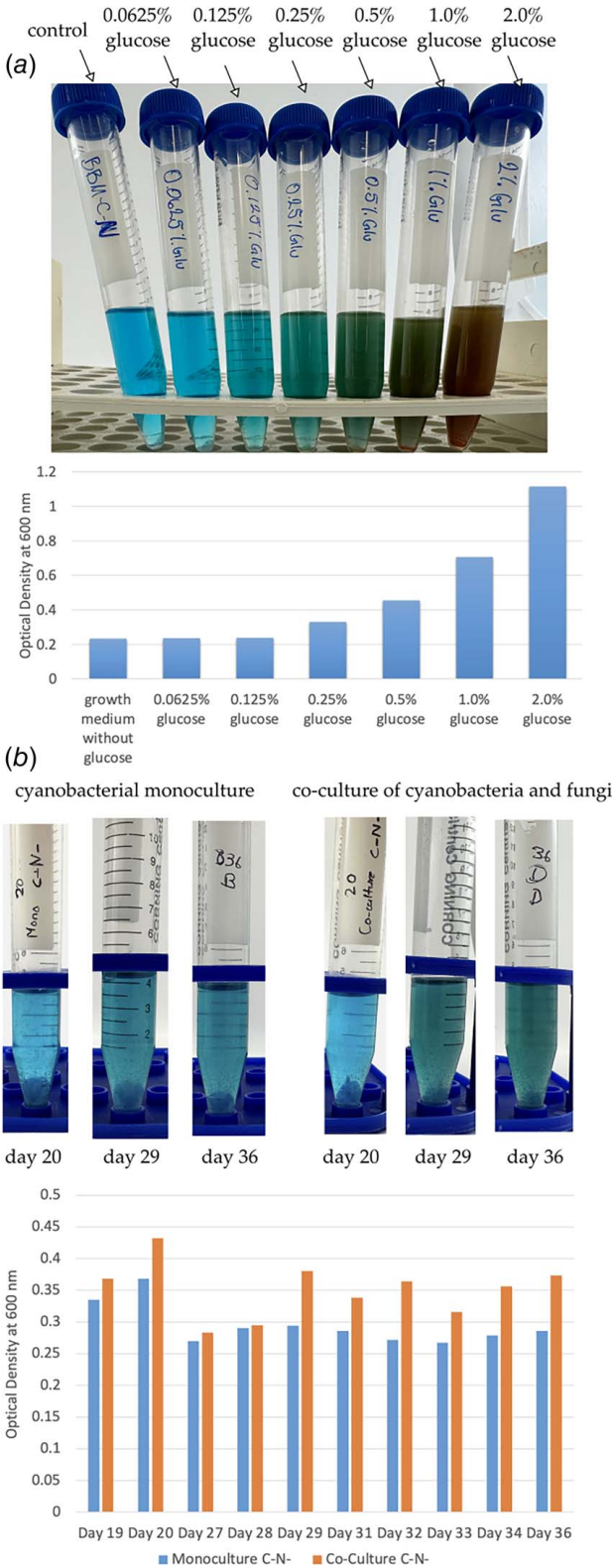


Fig. 9 Results of Benedict's test used to identify reducing sugars present in the supernatant of cyanobacterial monoculture wells and coculture wells, both grown solely on Martian regolith simulants, air, light, and an inorganic liquid medium without additional carbon or nitrogen sources

and morphologies of the precipitated crystals vary remarkably depending on the fungal component. The potential of this self-growing technology to enable long-term extraterrestrial exploration and colonization is significant.

The next step of the project, which is still ongoing, is to develop a regolith ink to print biostructures using the 3D printing technique of direct ink writing. The ink consists of cyanobacterial cells, fungal cells, and Martian regolith dispersed in the inorganic growth medium and its printability is optimized by alginate. Cyanobacteria readily produce alginate as an extracellular polymeric substance [24]. The physical crosslinking of alginate takes charge of initial regolith bonding to achieve self-supported printing, and the microbial cells generate biominerals for permanent regolith bonding.

The implementation has three major steps. As schematically illustrated in Fig. 11(a). The first step is to create a weak hydrogel using alginate and the calcium ions from Martian regolith simulants. To make the bioink, cyanobacterial cells, fungal cells, and Martian regolith simulants are dispersed in this inorganic growth medium. To improve the dispersion of alginate, the alginate powder is first mixed with water to create an alginate solution, which is added to the bioink right before printing. The ionic crosslinking of alginate with the calcium ions from

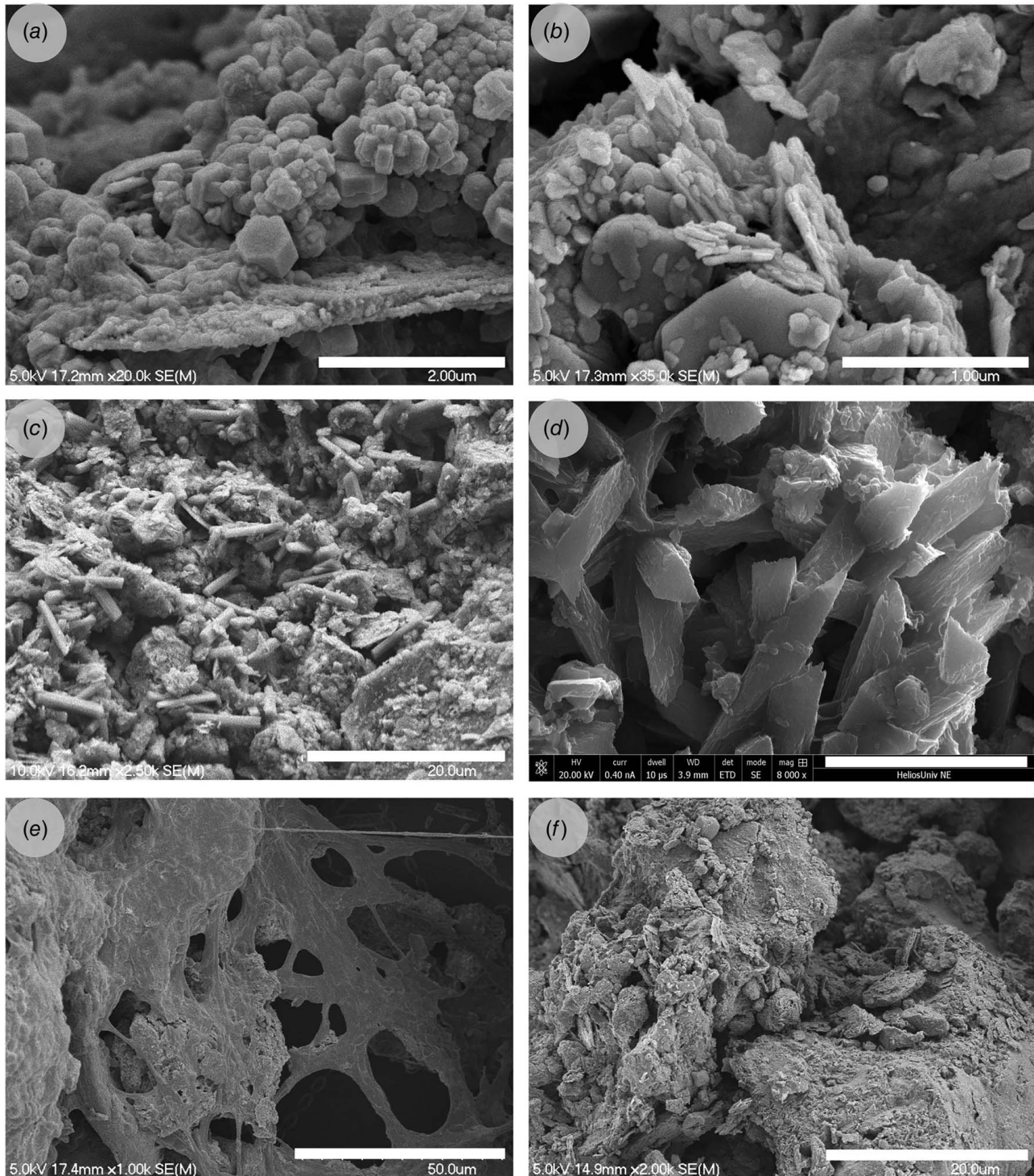


Fig. 10 Representative SEM images of precipitated crystals taken after 14-day incubation from the C – N– coculture wells whose fungal component is (a) *T. reesei*, (b) *T. viride* (HF3MP), (c) *T. viride* (HF3TV), (d) *P. sipitatus*, (e) *A. niger*, and (f) the microbe-free control growth medium. The scale bar is 2 μ m in (a), 1 μ m in (b), 20 μ m in (c), 10 μ m in (d), 50 μ m in (e), and 20 μ m in (f).

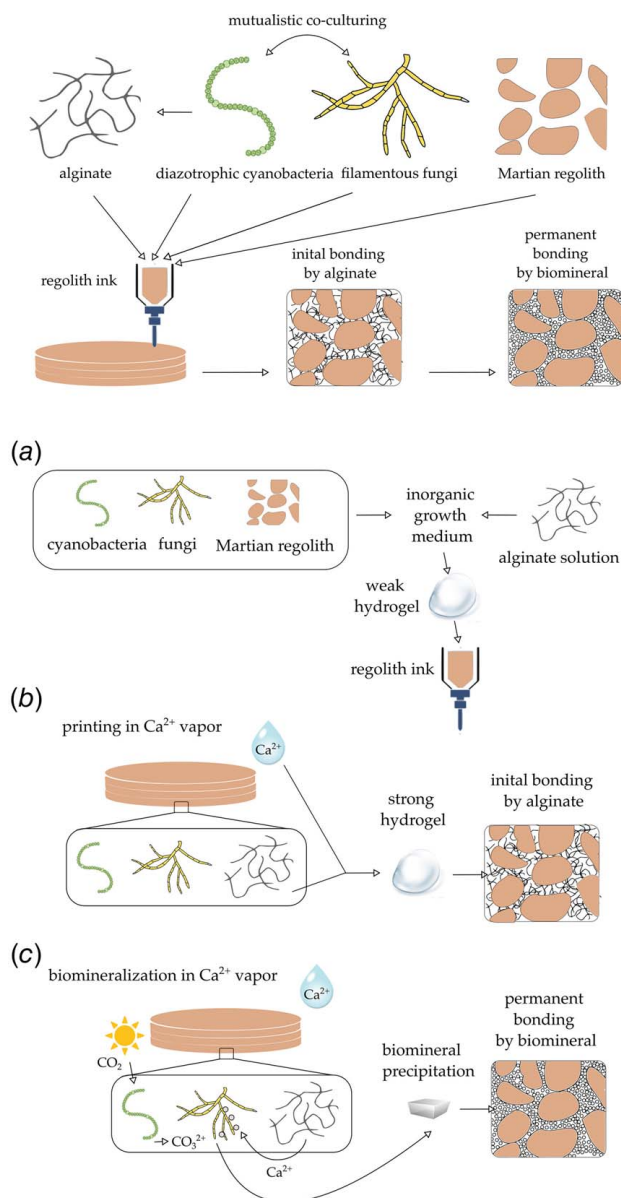


Fig. 11 The ongoing step is to develop a regolith ink to print biostructures using the 3D printing technique of direct ink writing. The physical crosslinking of alginate takes charge of initial regolith bonding to achieve self-supported printing, and the microbial cells generate biominerals for permanent regolith bonding.

Martian regolith simulants turns the inorganic growth medium into a weak hydrogel.

The weak hydrogel serves three roles, i.e., (1) to maintain homogeneity of material distribution, since the use of a gel-phase bioink was found to be able to prevent printer clogging [25]; (2) to provide cell protection during printing, since weak hydrogels provided more mechanical protection to encapsulated cells during clinically relevant cell transplantation, as compared to the case using a viscous fluid [26] and thus it is expected to be able to provide mechanical shielding during bioink printing; and (3) to maintain mechanical integrity of the printed structure, since weak hydrogels can still achieve excellent shape fidelity even when directly printed into an aqueous solution [27].

As shown in Fig. 11(b), the second step is to create a strong hydrogel for initial regolith bonding by printing in a vapor containing calcium ions [28]. The structure is directly printed in the vapor within the photobioreactor, which can be created by nebulizing a

solution containing calcium ions. The storage modulus of the alginate gel can be significantly increased by a moderate increase in the concentration of calcium ions [29]. This enhanced physical cross-linking of alginate takes charge of the initial regolith bonding to maintain the printed structure. This step also provides sufficient calcium ions to ensure sustained biomineral production and prevent cell dehydration.

As shown in Fig. 11(c), the third step is to create permanent regolith bonding using self-sustained production of biominerals. With an appropriate gelling degree, alginate hydrogel allows for cell-cell interaction and material transport within the porous matrix and provides a congenial environment for embedded cyanobacterial [27] and fungal [30] growth. The two species work together to generate biominerals for permanent regolith bonding. During biomineralization, alginate serves as a reservoir of calcium ions, facilitating localized production of calcium carbonate inside the printed structure. The mycelial network of the fungi serves as nucleation sites and strong framework support for biomineral precipitation. These fungal hyphae act as bridges connecting precipitated particles, thus reinforcing the formed biostructures. The biomineralization process can be completed within one month or terminated at any time when the microbes are killed by high temperatures or radiation.

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Conflict of Interest

There are no conflicts of interest.

Data Availability Statement

The datasets generated and supporting the findings of this article are obtainable from the corresponding author upon reasonable request.

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